Natural Language Processing Assignment 1: Interpretability of Model Collapse in Induction Heads

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Abstract

We study whether a small attention-only transformer trained on Singh et al.'s (2024) symbolic in-context-learning task forms a genuine induction circuit, and how that circuit and the model's output distribution behave under recursive self-training (model collapse; Shumailov et al. 2023). On the base (single-label) task, training shows a short ($\sim$12-epoch) phase of genuine shared train/validation improvement before a sharp overfitting cliff; the best checkpoint reaches 37% query-label test accuracy (chance: 25%) – a weak but real signal, not a clean circuit. The extended task (which also asks the model to generate a fresh symbol-label pair) never shows a shared-learning phase, and its dominant attention pattern is an attention-sink artifact rather than induction. Under an 8-generation recursive-retraining grid, deterministic sampling produces textbook distributional collapse – output entropy falls from 9.5 to 3.8 bits and up to 42% of continuations become structurally invalid – while stochastic sampling prevents this almost entirely. Embedding-geometry collapse shows the opposite pattern, indicating two distinct, dissociable failure modes.

Keywords: induction heads; in-context learning; model collapse

Introduction

In-context learning in transformers is often mediated by induction heads: a previous-token head feeding an induction head, together forming a circuit that completes patterns directly from context rather than from memorised weights (Olsson et al., 2022). Singh et al. (2024) study when and how such circuits form on a synthetic symbol-label task designed so in-weights memorisation cannot solve it. Separately, model collapse describes the degradation that occurs when generative models are trained recursively on their own outputs rather than real data (Shumailov et al., 2023).

This project asks two linked questions: does an induction circuit reliably form on Singh et al.'s task under a modest compute budget, and when the resulting model is recursively retrained on its own generations – using an extended task variant that gives the model's output distribution room to drift – does distributional and/or circuit collapse emerge, and if so, which happens first? We implement a base task (predict only the in-context query label) and an extended task (additionally generate one further, freshly-drawn symbol-label pair – a position unpredictable from context by construction, and therefore the natural lever for the model's own biases to compound under recursion), train an attention-only transformer on each, and subject the resulting models to an 8-generation recursive-retraining grid crossing data mode (pure/mixed) and sampling temperature (clean/noisy).

Methods

Task. Context sequences are (symbol, label) pairs under a randomised, per-sequence mapping, resampled every sequence. The base task scores only the query label; the extended task additionally scores one further, freshly-drawn pair.

Model. A 2-layer, 4-head attention-only transformer (no MLP sublayer, $d_{model} = 128$), following Olsson et al. (2022) and Singh et al. (2024) to avoid a competing memorisation pathway.

Training. A 1,000-symbol vocabulary with burstiness (k-shot $= 3$), following Chan et al. (2022)'s finding that vocabulary size and burstiness jointly favour in-context over in-weights solutions. Early stopping and checkpoint selection use a query-label-specific validation loss, since several scored positions (the extra symbol/label; repeated context-label positions) are by design unpredictable from context and would otherwise dilute the selection signal.

Collapse pipeline. Generation 0 trains on real data. Each subsequent generation (up to 8) trains on sequences whose context is always freshly real but whose scored continuation is the previous generation's own output. We cross *mode* (pure: 100% self-generated; mixed: 80%/20% real) with *sampling temperature* (0.0 greedy/"clean"; 1.0 ancestral/"noisy"), giving 4 extended-task conditions plus a base-task pure/noisy control.

Results

Baseline circuit formation. On the base task, train and validation accuracy rise together for $\sim$12 epochs before a sharp overfitting cliff: train accuracy reaches $\sim$100% while validation accuracy worsens. The best checkpoint reaches 37.0% query-label test accuracy (25% chance; $n_{ways} = 4$), with the strongest head scoring 0.077 on the induction metric – real but far short of a clean circuit, and no single head is critical under ablation. The extended task shows no shared learning phase at all; its dominant attention pattern is an attention sink rather than induction, and query-label accuracy (31.9%) is only slightly above chance.

Task accuracy under collapse. Across all four extended-task conditions, query-label accuracy over 8 generations is flat and noisy ($\sim$0.30–0.32), with no consistent trend. The base-task control declines mildly, from 0.32 (generation 0) to 0.29 (generation 7), despite the brief's own prediction that the base task should show little visible collapse.

Distributional collapse. This is the cleanest signal observed. Under deterministic sampling, the entropy of the model's own generated-symbol marginal collapses from 9.5 to 3.8 bits (pure) or 6.0 bits (mixed) by generation 7, and the

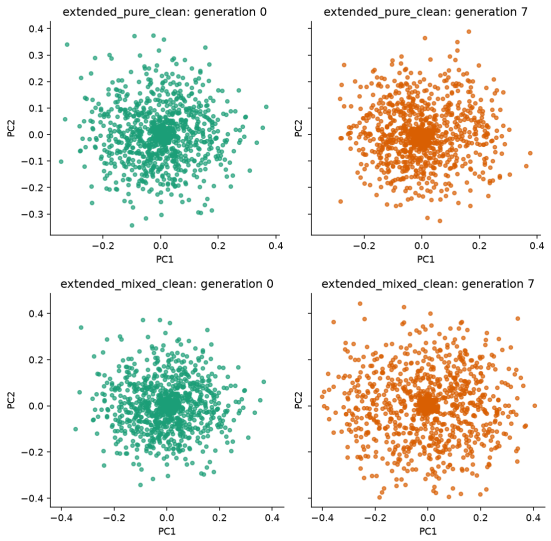


Figure 1: Symbol-embedding PCA (gen. 0 vs. 7), clean-sampling conditions. *pure_clean* (top, PR 41.9→42.0) and *mixed_clean* (bottom, PR 41.9→31.8) look identical despite only one collapsing – the 2D projection hides the signal participation ratio detects.

rate of structurally invalid outputs rises to 42% (pure) / 18–20% (mixed). Stochastic sampling shows almost no drift (entropy pinned at 9.2–9.3 bits; invalid rate $<$1%) throughout.

Representation and circuit metrics. Embedding participation ratio is flat for $\sim$5 generations, then drops sharply ($\sim$42 $\rightarrow$ $\sim$31) for three of the four conditions – *pure_noisy*, *mixed_noisy*, and *mixed_clean* – while *pure_clean* alone stays essentially flat (42.0 $\rightarrow$ 42.0); this is the opposite pattern from distributional collapse, where it is purity rather than sampling temperature that is protective. Strikingly, this collapse is invisible in a PC1/PC2 projection of the embeddings (Figure 1): *pure_clean* and *mixed_clean* look equally diffuse at generation 7 despite one collapsing in participation ratio and the other not, indicating the collapse is distributed across higher-order components rather than the two dominant directions. The induction score of the strongest head *rises* over generations in most conditions even as functional accuracy does not improve, showing this attention-based proxy does not reliably track task performance.

Discussion

The circuit that forms here is weak and partial rather than the clean circuit reported in idealised settings: even with the vocabulary/burstiness mitigations from Chan et al. (2022), a competing in-weights pathway overtakes the nascent inductive solution around epoch 12.

The central finding is that distributional collapse (in the extended task's freely-generated symbol) and representation collapse (embedding geometry) are dissociable rather than one phenomenon. Distributional collapse is driven almost entirely by sampling temperature, consistent with standard model-collapse theory (Shumailov et al., 2023): purity does not matter, only whether sampling is deterministic. Representation collapse inverts this logic – there, purity is what protects (only *pure_clean* survives intact), and temperature does not – and it is detectable only via the embedding spectrum (participation ratio), not via a low-dimensional visual check. Because the baseline circuit is already weak, we cannot cleanly isolate whether *circuit* collapse specifically tracks either signal under a stronger starting circuit.

Limitations. Single run per condition; modest compute scale; the induction-score/accuracy decoupling warrants a causal check (e.g. activation patching) rather than attention statistics alone. **Future work:** multiple seeds, a stronger starting circuit, and generation-by-generation causal measurement of circuit health.

Contribution Statements (Not in Page Limit)

Both authors worked independently on the project before combining their respective implementations, experiments, and findings into the final submission. Individual responsibilities differed – Salimata Mbaye led the submitted implementation, ran the baseline and collapse-grid experiments, conducted the interpretability analyses, and prepared this abstract; Mashego Mabeloane independently developed and investigated the project implementation and experimental workflow, including diagnosing baseline training issues and running the baseline and collapse-grid experiments, and contributed to result interpretation – but both authors agree the overall contribution was divided equally (50%/50%). Both authors confirm this statement is accurate and agree to its submission.

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